

CHSH Game

$$R_y(\theta) = \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}, H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$|\psi_0\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

$$(0,0) \rightarrow \begin{bmatrix} \cos(\frac{\pi}{8}) & -\sin(\frac{\pi}{8}) \\ \sin(\frac{\pi}{8}) & \cos(\frac{\pi}{8}) \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

$$= \begin{bmatrix} \cos(\frac{\pi}{8}) & 0 & \sin(\frac{\pi}{8}) & 0 \\ 0 & \cos(\frac{\pi}{8}) & 0 & \sin(\frac{\pi}{8}) \\ -\sin(\frac{\pi}{8}) & 0 & \cos(\frac{\pi}{8}) & 0 \\ 0 & -\sin(\frac{\pi}{8}) & 0 & \cos(\frac{\pi}{8}) \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} \cos(\frac{\pi}{8}) \\ \sin(\frac{\pi}{8}) \\ -\sin(\frac{\pi}{8}) \\ \cos(\frac{\pi}{8}) \end{bmatrix}$$

$\therefore$ To win: $|00\rangle, |11\rangle$

$$\therefore \text{Prob win} = \left(\frac{1}{\sqrt{2}} \cos\left(\frac{\pi}{8}\right) \right)^2 + \left(\frac{1}{\sqrt{2}} \cos\left(\frac{\pi}{8}\right) \right)^2 = \cos^2\left(\frac{\pi}{8}\right)$$

$$(0,1) \rightarrow \begin{bmatrix} \cos(\frac{\pi}{8}) & 0 & -\sin(\frac{\pi}{8}) & 0 \\ 0 & \cos(\frac{\pi}{8}) & 0 & -\sin(\frac{\pi}{8}) \\ \sin(\frac{\pi}{8}) & 0 & \cos(\frac{\pi}{8}) & 0 \\ 0 & \sin(\frac{\pi}{8}) & 0 & \cos(\frac{\pi}{8}) \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} \cos(\frac{\pi}{8}) \\ -\sin(\frac{\pi}{8}) \\ \sin(\frac{\pi}{8}) \\ \cos(\frac{\pi}{8}) \end{bmatrix}$$

$\therefore$ To win: $|00\rangle, |11\rangle$

$$\therefore \text{Prob win} = \left(\frac{1}{\sqrt{2}} \cos\left(\frac{\pi}{8}\right) \right)^2 + \left(\frac{1}{\sqrt{2}} \cos\left(\frac{\pi}{8}\right) \right)^2 = \cos^2\left(\frac{\pi}{8}\right)$$

$$(1,0) \rightarrow \begin{bmatrix} \cos(\frac{\pi}{8}) & -\sin(\frac{\pi}{8}) \\ \sin(\frac{\pi}{8}) & \cos(\frac{\pi}{8}) \end{bmatrix} \otimes \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

$$= \frac{1}{2} \begin{bmatrix} \cos(\frac{\pi}{8}) & \cos(\frac{\pi}{8}) & \sin(\frac{\pi}{8}) & \sin(\frac{\pi}{8}) \\ \cos(\frac{\pi}{8}) & -\cos(\frac{\pi}{8}) & \sin(\frac{\pi}{8}) & -\sin(\frac{\pi}{8}) \\ -\sin(\frac{\pi}{8}) & -\sin(\frac{\pi}{8}) & \cos(\frac{\pi}{8}) & \cos(\frac{\pi}{8}) \\ -\sin(\frac{\pi}{8}) & \sin(\frac{\pi}{8}) & \cos(\frac{\pi}{8}) & -\cos(\frac{\pi}{8}) \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} \cos(\frac{\pi}{8}) + \sin(\frac{\pi}{8}) \\ \cos(\frac{\pi}{8}) - \sin(\frac{\pi}{8}) \\ -\sin(\frac{\pi}{8}) + \cos(\frac{\pi}{8}) \\ -\sin(\frac{\pi}{8}) - \cos(\frac{\pi}{8}) \end{bmatrix}$$

$\therefore$ To win: $|00\rangle, |11\rangle$

$$\begin{aligned} \therefore \text{Prob win} &= \left(\frac{1}{2} (\cos(\frac{\pi}{8}) + \sin(\frac{\pi}{8})) \right)^2 + \left(\frac{1}{2} (\cos(\frac{\pi}{8}) - \sin(\frac{\pi}{8})) \right)^2 \\ &= \frac{1}{2} \left(\cancel{\cos^2(\frac{\pi}{8})} + 2\cos(\frac{\pi}{8})\sin(\frac{\pi}{8}) + \cancel{\sin^2(\frac{\pi}{8})} \right) \\ &= \frac{1}{2} \left(1 + 2\cos(\frac{\pi}{8})\sin(\frac{\pi}{8}) \right) \end{aligned}$$

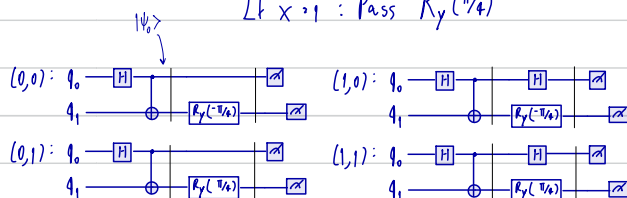
Strategy: $|q_1 q_0\rangle, (x, y)$

To win: $x \wedge y = a \oplus b$

x	y	win
0	0	$a=b$
0	1	$a=b$
1	0	$a=b$
1	1	$a \neq b$

For q_0 : If $x=0$: Measure with basis Z (No gates)
If $x=1$: Measure with basis X (Pass H)

For q_1 : If $x=0$: Pass $R_y(-\pi/4)$
If $x=1$: Pass $R_y(\pi/4)$



$$= \frac{1}{2} + \cos\left(\frac{\pi}{8}\right) \sin\left(\frac{\pi}{8}\right)$$

$$= \cos^2\left(\frac{\pi}{8}\right) \left(\frac{1}{2 \cos^2\left(\frac{\pi}{8}\right)} + \frac{\sin\left(\frac{\pi}{8}\right)}{\cos\left(\frac{\pi}{8}\right)} \right)$$

$$= \cos^2\left(\frac{\pi}{8}\right) \left(\frac{1}{2 \cancel{\cos^2\left(\frac{\pi}{8}\right)}} + \sqrt{2} - 1 \right)$$

$$= \cos^2\left(\frac{\pi}{8}\right) \left(\frac{2}{2+\sqrt{2}} + \frac{(\sqrt{2}-1)(2+\sqrt{2})}{2+\sqrt{2}} \right)$$

$$= \cos^2\left(\frac{\pi}{8}\right) \left(\frac{\cancel{2} + \cancel{2}\sqrt{2} + \cancel{2} - \cancel{2}}{\cancel{2} + \sqrt{2}} \right)$$

$$= \cos^2\left(\frac{\pi}{8}\right) (1)$$

$$= \cos^2\left(\frac{\pi}{8}\right)$$

$$(1,1) \rightarrow \frac{1}{2} \begin{bmatrix} \cos\left(\frac{\pi}{8}\right) & \cos\left(\frac{\pi}{8}\right) & -\sin\left(\frac{\pi}{8}\right) & -\sin\left(\frac{\pi}{8}\right) \\ \cos\left(\frac{\pi}{8}\right) & -\cos\left(\frac{\pi}{8}\right) & -\sin\left(\frac{\pi}{8}\right) & \sin\left(\frac{\pi}{8}\right) \\ \sin\left(\frac{\pi}{8}\right) & \sin\left(\frac{\pi}{8}\right) & \cos\left(\frac{\pi}{8}\right) & \cos\left(\frac{\pi}{8}\right) \\ \sin\left(\frac{\pi}{8}\right) & -\sin\left(\frac{\pi}{8}\right) & \cos\left(\frac{\pi}{8}\right) & -\cos\left(\frac{\pi}{8}\right) \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} \cos\left(\frac{\pi}{8}\right) - \sin\left(\frac{\pi}{8}\right) \\ \cos\left(\frac{\pi}{8}\right) + \sin\left(\frac{\pi}{8}\right) \\ \sin\left(\frac{\pi}{8}\right) + \cos\left(\frac{\pi}{8}\right) \\ \sin\left(\frac{\pi}{8}\right) - \cos\left(\frac{\pi}{8}\right) \end{bmatrix}$$

$$\therefore \text{To win: } |01\rangle, |10\rangle$$

$$\therefore \text{Prob win} = \left(\frac{1}{2} (\cos\left(\frac{\pi}{8}\right) - \sin\left(\frac{\pi}{8}\right)) \right)^2 + \left(\frac{1}{2} (\sin\left(\frac{\pi}{8}\right) + \cos\left(\frac{\pi}{8}\right)) \right)^2$$

$$\vdots$$

same as (1,0)

$$= \cos^2\left(\frac{\pi}{8}\right)$$

$$\therefore \text{Prob of win total} = \frac{1}{4} \cos^2\left(\frac{\pi}{8}\right) + \frac{1}{4} \cos^2\left(\frac{\pi}{8}\right) + \frac{1}{4} \cos^2\left(\frac{\pi}{8}\right) + \frac{1}{4} \cos^2\left(\frac{\pi}{8}\right)$$

(assume x, y are uniformly distributed)

$$= \cos^2\left(\frac{\pi}{8}\right)$$

$$\approx 0.85$$



20 2
1 10 3
5 2